

MATHEMATICS STANDARD 2026 ALL SETS

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CHAPTER: 1 - Real Numbers

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YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $\sqrt{0.4}$ is a/an

OPTIONS: (A) natural number (B) integer (C) rational number (D) irrational number

ANSWER: (D) irrational number

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following cannot be the unit digit of $8n$, where n is a natural number?

OPTIONS: (A) 4 (B) 2 (C) 0 (D) 6

ANSWER: (C) 0

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Let $x = a^2b^3c^n$ and $y = a^3b^mc^2$, where a, b, c are prime numbers. If LCM of x and y is $a^3b^4c^3$, then the value of $m + n$ is

OPTIONS: (A) 10 (B) 7 (C) 6 (D) 5

ANSWER: (B) 7

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $1080 = 2^p \times 3^q \times 5$, then $(p - q)$ is equal to:

OPTIONS: (A) 6 (B) -1 (C) 1 (D) 0

ANSWER: (D) 0

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The least number which is a perfect square and is divisible by each of 16, 20 and 50, is:

OPTIONS: (A) 1200 (B) 100 (C) 3600 (D) 2400

ANSWER: The correct option is not available in the given options. Full marks may be awarded to every attempt.

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $(1 + \sqrt{3})^2 - (1 - \sqrt{3})^2$ is:

OPTIONS: (A) a positive rational number (B) a negative integer (C) a positive irrational number (D) a negative irrational number

ANSWER: (C) a positive irrational number

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $x = ab^3$ and $y = a^3b$, where a and b are prime numbers, then $[HCF(x, y) - LCM(x, y)]$ is equal to:

OPTIONS: (A) $1 - a^3b^3$ (B) $ab(1 - ab)$ (C) $ab - a^4b^4$ (D) $ab(1 - ab)(1 + ab)$

ANSWER: (D) $ab(1 - ab)(1 + ab)$

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The sum of the exponents of prime factors in the prime factorisation of 4004 is:

OPTIONS: (A) 5 (B) 4 (C) 3 (D) 2

ANSWER: (A) 5

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\text{HCF}(98, 28) = m$ and $\text{LCM}(98, 28) = n$, then the value of $n - 7m$ is:

OPTIONS: (A) 0 (B) 28 (C) 98 (D) 198

ANSWER: (C) 98

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Let $a = p^2q^3r^m$ and $b = p^3q^mr^2$, where p, q, r are prime numbers. If LCM of a and b is $p^3q^4r^3$, then the value of $3n - 2m$ is

OPTIONS: (A) -1 (B) 1 (C) 3 (D) -3

ANSWER: (B) 1

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The HCF of 40, 110 and 360 is:

OPTIONS: (A) 40 (B) 110 (C) 360 (D) 10

ANSWER: (D) 10

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For any prime number p , if p divides a^2 , where a is any real number then p also divides

OPTIONS: (A) a (B) a^2 (C) a^3 (D) a^5

ANSWER: (A) a

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $a^b = 32$, where 'a' and 'b' are positive integers, then the value of b^{ab} is:

OPTIONS: (A) 72 (B) 5^{10} (C) 2^{10} (D) 5^{12}

ANSWER: (B) 5^{10}

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the length of the plank that can be used to measure the lengths 4 m 20 cm and 5 m 4 cm exactly, in the least time.

OPTIONS: (A) 84 cm (B) 72 cm (C) 96 cm (D) 108 cm

ANSWER: (A) 84 cm

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Three sets of Physics, Chemistry and Mathematics books have to be stacked in such a way that all the books are stored subject-wise and the height of each stack is the same. The number of Physics books is 144, the number of Chemistry books is 180 and the number of Mathematics books is 192. Assuming that the books are of same thickness, determine the number of stacks of Physics, Chemistry and Mathematics books.

OPTIONS: (A) 12, 15, 16 (B) 10, 12, 14 (C) 15, 18, 20 (D) 12, 14, 18

ANSWER: (A) 12, 15, 16

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The greatest number which divides 70 and 125, leaving remainders 5 and 8 respectively, is:

OPTIONS: (A) 13 (B) 65 (C) 875 (D) 1750

ANSWER: (A) 13

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If the HCF of 210 and 55 is expressed as $210 \times 5 + 55m$, then find the value of m .

OPTIONS: (A) -19 (B) -17 (C) 19 (D) 17

ANSWER: (A) -19

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the smallest number which is divisible by both 644 and 462.

OPTIONS: (A) 21252 (B) 10626 (C) 42504 (D) 5313

ANSWER: (A) 21252

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Two numbers are in the ratio 4 : 5 and their HCF is 11. Find the LCM of these numbers.

OPTIONS: (A) 220 (B) 110 (C) 440 (D) 55

ANSWER: (A) 220

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Let p , q and r be three distinct prime numbers. Check whether $p \cdot q \cdot r + q$ is a composite number or not. Further, give an example for 3 distinct primes p , q , r such that (i) $p \cdot q \cdot r + 1$ is a composite number. (ii) $p \cdot q \cdot r + 1$ is a prime number.

OPTIONS: (A) Composite; $p=3, q=5, r=7$ gives 106 (composite); $p=2, q=3, r=5$ gives 31 (prime) (B) Prime; $p=3, q=5, r=7$ gives 106 (prime); $p=2, q=3, r=5$ gives 31 (composite) (C) Composite;

$p=3, q=5, r=7$ gives 106 (prime); $p=2, q=3, r=5$ gives 31 (composite) (D) Prime; $p=3, q=5, r=7$ gives 106 (composite); $p=2, q=3, r=5$ gives 31 (prime)

ANSWER: (A) Composite; $p=3, q=5, r=7$ gives 106 (composite); $p=2, q=3, r=5$ gives 31 (prime)

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: State true or false for each of the following statements and justify in each case: (i) $2 \times 3 \times 5 \times 7 + 7$ is a composite number. (ii) $2 \times 3 \times 5 \times 7 + 1$ is a composite number.

OPTIONS: (A) (i) True, (ii) False (B) (i) False, (ii) True (C) Both True (D) Both False

ANSWER: (A) (i) True, (ii) False

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Let x and y be two distinct prime numbers and $p = x^2y^3$, $q = xy^4$, $r = x^5y^2$. Find the HCF and LCM of p , q and r . Further check if $\text{HCF}(p, q, r) \times \text{LCM}(p, q, r) = p \times q \times r$ or not.

OPTIONS: (A) $\text{HCF} = xy^2$, $\text{LCM} = x^5y^4$, $\text{HCF} \times \text{LCM} \neq p \times q \times r$ (B) $\text{HCF} = xy^2$, $\text{LCM} = x^5y^4$, $\text{HCF} \times \text{LCM} = p \times q \times r$ (C) $\text{HCF} = x^2y^2$, $\text{LCM} = x^5y^4$, $\text{HCF} \times \text{LCM} \neq p \times q \times r$ (D) $\text{HCF} = xy^2$, $\text{LCM} = x^5y^3$, $\text{HCF} \times \text{LCM} \neq p \times q \times r$

ANSWER: (A) $\text{HCF} = xy^2$, $\text{LCM} = x^5y^4$, $\text{HCF} \times \text{LCM} \neq p \times q \times r$

CHAPTER 2 - POLYNOMIALS

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CHAPTER: 2 - Polynomials

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YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following statements is true for a polynomial $p(x)$ of degree 3?

OPTIONS: (A) $p(x)$ has at most two distinct zeroes. (B) $p(x)$ has at least two distinct zeroes. (C) $p(x)$ has exactly three distinct zeroes. (D) $p(x)$ has at most three distinct zeroes.

ANSWER: (D) $p(x)$ has at most three distinct zeroes.

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If -4 is a zero of the polynomial $p(x) = x^2 - x - (2 + 2k)$, then the value of k is:

OPTIONS: (A) 3 (B) 9 (C) 6 (D) -9

ANSWER: (B) 9

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the zeroes of the polynomial $ax^2 + bx + 2a/b$ are reciprocal of each other, then the value of b is

OPTIONS: (A) 2 (B) 1/2 (C) -2 (D) -1/2

ANSWER: (A) 2

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If α and β are the zeroes of the polynomial $p(x) = x^2 - ax - b$, then the value of $(\alpha + \beta + \alpha\beta)$ is equal to:

OPTIONS: (A) $a + b$ (B) $-a - b$ (C) $a - b$ (D) $-a + b$

ANSWER: (C) $a - b$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Zeroes of the polynomial $p(x) = x^2 - 3\sqrt{2}x + 4$ are:

OPTIONS: (A) 2, $\sqrt{2}$ (B) $2\sqrt{2}$, $\sqrt{2}$ (C) $4\sqrt{2}$, $-\sqrt{2}$ (D) $\sqrt{2}$, 2

ANSWER: (B) $2\sqrt{2}$, $\sqrt{2}$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If one zero of the polynomial $q(x) = (p^2 + 4)x^2 + 65x + 4p$ is reciprocal of the other, then the value of 'p' is:

OPTIONS: (A) -1 (B) 1 (C) -2 (D) 2

ANSWER: (D) 2

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The sum of the zeroes of the polynomial $p(x) = 5x - 7x^2 + 3$ is:

OPTIONS: (A) -7/5 (B) 7/5 (C) 5/7 (D) -5/7

ANSWER: (C) 5/7

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the zeroes of the polynomial $p(x) = x^2 + 4/3x - 4/3$.

OPTIONS: (A) 2/3, -2 (B) -2/3, 2 (C) 2/3, 2 (D) -2/3, -2

ANSWER: (A) 2/3, -2

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If the zeroes of the polynomial $x^2 + ax + b$ are in the ratio 3 : 4, then prove that $12a^2 = 49b$.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If p and q are zeroes of the polynomial $p(y) = 21y^2 - y - 2$, then find the value of $(1 - p) \cdot (1 - q)$.

OPTIONS: (A) $6/7$ (B) $7/6$ (C) $3/7$ (D) $2/7$

ANSWER: (A) $6/7$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find a quadratic polynomial whose zeroes are 2 and $-7/5$.

OPTIONS: (A) $5x^2 - 3x - 14$ (B) $5x^2 + 3x - 14$ (C) $5x^2 - 3x + 14$ (D) $5x^2 + 3x + 14$

ANSWER: (A) $5x^2 - 3x - 14$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If the sum of the zeroes of the polynomial $p(x) = (p + 1)x^2 + (2p + 3)x + (3p + 4)$ is -1 , then find the value of ' p '.

OPTIONS: (A) -2 (B) 2 (C) -1 (D) 1

ANSWER: (A) -2

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α and β are zeroes of the polynomial $p(x) = x^2 - 2x - 1$, then find the value of $1/(2\alpha) + 1/(2\beta) + 3\alpha\beta$.

OPTIONS: (A) -4 (B) 4 (C) -2 (D) 2

ANSWER: (A) -4

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Obtain the zeroes of the polynomial $p(x) = 2x^2 - 5x - 3$. Hence, obtain a polynomial each of whose zeroes is one less than each of the zero of $p(x)$.

OPTIONS: (A) Zeroes: 3, $-1/2$; New polynomial: $2x^2 - x - 6$ (B) Zeroes: -3 , $1/2$; New polynomial: $2x^2 - x - 6$ (C) Zeroes: 3, $-1/2$; New polynomial: $2x^2 + x - 6$ (D) Zeroes: -3 , $1/2$; New polynomial: $2x^2 + x - 6$

ANSWER: (A) Zeroes: 3, $-1/2$; New polynomial: $2x^2 - x - 6$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $p(x) = 3x^2 - 4x - 4$. Hence, write a polynomial whose each of the zeroes is 2 more than zeroes of $p(x)$.

OPTIONS: (A) Zeroes: $-2/3$, 2; New polynomial: $3x^2 - 16x + 16$ (B) Zeroes: $2/3$, -2 ; New polynomial: $3x^2 + 16x + 16$ (C) Zeroes: $-2/3$, 2; New polynomial: $3x^2 + 16x + 16$ (D) Zeroes: $2/3$, -2 ; New polynomial: $3x^2 - 16x + 16$

ANSWER: (A) Zeroes: $-2/3$, 2; New polynomial: $3x^2 - 16x + 16$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Obtain the zeroes of the polynomial $7x^2 + 18x - 9$. Hence, write a polynomial each of whose zeroes is twice the zeroes of given polynomial.

OPTIONS: (A) Zeroes: -3 , $3/7$; New polynomial: $7x^2 + 12x - 9$ (B) Zeroes: 3, $-3/7$; New polynomial: $7x^2 + 12x - 9$ (C) Zeroes: -3 , $3/7$; New polynomial: $7x^2 - 12x - 9$ (D) Zeroes: 3, $-3/7$; New polynomial: $7x^2 - 12x - 9$

ANSWER: (A) Zeroes: -3 , $3/7$; New polynomial: $7x^2 + 12x - 9$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $q(x) = 8x^2 - 2x - 3$. Hence, find a polynomial whose zeroes are 2 less than the zeroes of $q(x)$.

OPTIONS: (A) Zeroes: $-\frac{1}{2}$, $\frac{3}{4}$; New polynomial: $8x^2 + 30x + 25$ (B) Zeroes: $\frac{1}{2}$, $-\frac{3}{4}$; New polynomial: $8x^2 - 30x + 25$ (C) Zeroes: $-\frac{1}{2}$, $\frac{3}{4}$; New polynomial: $8x^2 - 30x + 25$ (D) Zeroes: $\frac{1}{2}$, $-\frac{3}{4}$; New polynomial: $8x^2 + 30x + 25$
ANSWER: (A) Zeroes: $-\frac{1}{2}$, $\frac{3}{4}$; New polynomial: $8x^2 + 30x + 25$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $p(x) = 6x^2 - 5x - 1$. Hence, obtain a polynomial each of whose zeroes is three times the zeroes of $p(x)$.

OPTIONS: (A) Zeroes: 1, $-\frac{1}{6}$; New polynomial: $2x^2 - 5x - 3$ (B) Zeroes: -1, $\frac{1}{6}$; New polynomial: $2x^2 - 5x - 3$ (C) Zeroes: 1, $-\frac{1}{6}$; New polynomial: $2x^2 + 5x - 3$ (D) Zeroes: -1, $\frac{1}{6}$; New polynomial: $2x^2 + 5x - 3$

ANSWER: (A) Zeroes: 1, $-\frac{1}{6}$; New polynomial: $2x^2 - 5x - 3$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $r(x) = 4x^2 + 3x - 1$. Hence, write a polynomial whose zeroes are reciprocal of the zeroes of polynomial $r(x)$.

OPTIONS: (A) Zeroes: $\frac{1}{4}$, -1; New polynomial: $x^2 - 3x - 4$ (B) Zeroes: $-\frac{1}{4}$, 1; New polynomial: $x^2 + 3x - 4$ (C) Zeroes: $\frac{1}{4}$, -1; New polynomial: $x^2 + 3x - 4$ (D) Zeroes: $-\frac{1}{4}$, 1; New polynomial: $x^2 - 3x - 4$

ANSWER: (A) Zeroes: $\frac{1}{4}$, -1; New polynomial: $x^2 - 3x - 4$

CHAPTER 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

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CHAPTER: 3 - Pair of Linear Equations in Two Variables

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YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of 'k' for which the system of linear equations $6x + y = 3k$ and $36x + 6y = 3$ have infinitely many solutions is:

OPTIONS: (A) 6 (B) $\frac{1}{6}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

ANSWER: (B) $\frac{1}{6}$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A system of two linear equations in two variables is inconsistent, if the lines in the graph are:

OPTIONS: (A) coincident (B) parallel (C) intersecting at one point (D) intersecting at right angles

ANSWER: (B) parallel

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve the following system of equations algebraically: $30x + 44y = 10$; $40x + 55y = 13$

OPTIONS: (A) $x = \frac{1}{5}$, $y = \frac{1}{11}$ (B) $x = \frac{1}{5}$, $y = \frac{1}{10}$ (C) $x = \frac{1}{4}$, $y = \frac{1}{11}$ (D) $x = \frac{1}{4}$, $y = \frac{1}{10}$

ANSWER: (A) $x = \frac{1}{5}$, $y = \frac{1}{11}$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve the following system of equations algebraically: $37x + 63y = 137$; $63x + 37y = 163$

OPTIONS: (A) $x = 2$, $y = 1$ (B) $x = 1$, $y = 2$ (C) $x = 3$, $y = 1$ (D) $x = 1$, $y = 3$

ANSWER: (A) $x = 2$, $y = 1$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve the following system of equations algebraically: $73x - 37y = 109$; $37x - 73y = 1$

OPTIONS: (A) $x = 2, y = 1$ (B) $x = 1, y = 2$ (C) $x = 3, y = 2$ (D) $x = 2, y = 3$

ANSWER: (A) $x = 2, y = 1$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve the following pair of equations algebraically: $101x + 102y = 304$; $102x + 101y = 305$

OPTIONS: (A) $x = 2, y = 1$ (B) $x = 1, y = 2$ (C) $x = 3, y = 1$ (D) $x = 1, y = 3$

ANSWER: (A) $x = 2, y = 1$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In a pair of supplementary angles, the greater angle exceeds the smaller by 50° . Express the given situation as a system of linear equations in two variables and hence obtain the measure of each angle.

OPTIONS: (A) 65° and 115° (B) 60° and 120° (C) 70° and 110° (D) 55° and 125°

ANSWER: (A) 65° and 115°

YEAR: 2025-06

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The cost of 2 kg apples and 1 kg of grapes on a day was found to be ₹ 320. The cost of 4 kg apples and 2 kg grapes was found to be ₹ 600. If cost of 1 kg of apples and 1 kg of grapes is ₹ x and ₹ y respectively, represent the given situation algebraically as a system of equations and check whether the system so obtained is consistent or not.

OPTIONS: (A) $2x + y = 320, 4x + 2y = 600$; Inconsistent (B) $2x + y = 320, 4x + 2y = 600$; Consistent (C) $2x + y = 600, 4x + 2y = 320$; Inconsistent (D) $2x + y = 600, 4x + 2y = 320$; Consistent

ANSWER: (A) $2x + y = 320, 4x + 2y = 600$; Inconsistent

YEAR: 2025-06

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y: $\sqrt{2}x + \sqrt{3}y = 5$ and $\sqrt{3}x - \sqrt{8}y = -\sqrt{6}$

OPTIONS: (A) $x = \sqrt{2}$, $y = \sqrt{3}$ (B) $x = \sqrt{3}$, $y = \sqrt{2}$ (C) $x = 2$, $y = 3$ (D) $x = 3$, $y = 2$

ANSWER: (A) $x = \sqrt{2}$, $y = \sqrt{3}$

YEAR: 2025-06

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The monthly incomes of two persons are in the ratio 9 : 7 and their monthly expenditures are in the ratio 4 : 3. If each saved ₹ 5,000, express the given situation algebraically as a system of linear equations in two variables. Hence, find their respective monthly incomes.

OPTIONS: (A) ₹ 45,000 and ₹ 35,000 (B) ₹ 40,000 and ₹ 30,000 (C) ₹ 50,000 and ₹ 40,000 (D) ₹ 35,000 and ₹ 25,000

ANSWER: (A) ₹ 45,000 and ₹ 35,000

CHAPTER 4 - QUADRATIC EQUATIONS

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CHAPTER: 4 - Quadratic Equations

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YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The quadratic equation whose roots are 7 and $\frac{1}{7}$ is:

OPTIONS: (A) $7x^2 - 50x + 7 = 0$ (B) $7x^2 - 50x + 1 = 0$ (C) $7x^2 + 50x - 7 = 0$ (D) $7x^2 + 50x - 1 = 0$

ANSWER: (A) $7x^2 - 50x + 7 = 0$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The quadratic equation whose sum and product of roots are 'a' and '1/a' respectively is:

OPTIONS: (A) $ax^2 - ax + 1 = 0$ (B) $ax^2 - a^2x + 1 = 0$ (C) $ax^2 + ax + 1 = 0$ (D) $ax^2 + a^2x - 1 = 0$

ANSWER: (B) $ax^2 - a^2x + 1 = 0$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of 'a' for which $ax^2 + x + a = 0$ has equal and positive roots is:

OPTIONS: (A) 2 (B) -2 (C) 1/2 (D) -1/2

ANSWER: (D) -1/2

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following equations is a quadratic equation?

OPTIONS: (A) $x^2 + 1 = (x - 1)^2$ (B) $(x + \sqrt{x})^2 = 2x\sqrt{x}$ (C) $x^3 + 3x^2 = (x + 1)^3$ (D) $(x + 1)(x - 1) = (x + 1)^2$

ANSWER: (B) $(x + \sqrt{x})^2 = 2x\sqrt{x}$

YEAR: 2025-06

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $x^2 + bx + b = 0$ has two real and distinct roots, then the value of b can be

OPTIONS: (A) 0 (B) 4 (C) 3 (D) -3

ANSWER: (D) -3

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $x/12 - 3/x = 0$, then the values of x are:

OPTIONS: (A) ± 6 (B) ± 4 (C) ± 12 (D) ± 3

ANSWER: (A) ± 6

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following quadratic equations has real and equal roots?

OPTIONS: (A) $(x + 1)^2 = 2x + 1$ (B) $x^2 + x = 0$ (C) $x^2 - 4 = 0$ (D) $x^2 + x + 1 = 0$

ANSWER: (A) $(x + 1)^2 = 2x + 1$

YEAR: 2025-06

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following equations does not have a real root?

OPTIONS: (A) $x^2 = 0$ (B) $2x - 1 = 3$ (C) $x^2 + 1 = 0$ (D) $x^3 + x^2 = 0$

ANSWER: (C) $x^2 + 1 = 0$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the value(s) of 'k' so that the quadratic equation $4x^2 + kx + 1 = 0$ has real and equal roots.

OPTIONS: (A) ± 4 (B) ± 2 (C) ± 8 (D) ± 16

ANSWER: (A) ± 4

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If ' α ' and ' β ' are the zeroes of the polynomial $p(y) = y^2 - 5y + 3$, then find the value of $\alpha^4\beta^3 + \alpha^3\beta^4$.

OPTIONS: (A) 135 (B) 125 (C) 145 (D) 155

ANSWER: (A) 135

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A garden designer is planning a rectangular lawn that is to be surrounded by a uniform walkway. The total area of the lawn and the walkway is 360 square metres. The width of the walkway is same on all sides. The dimensions of the lawn itself are 12 metres by 10 metres. Formulate the quadratic equation representing the total area of the lawn and the walkway, taking width of walkway = x m. Solve the quadratic equation to find the width of the walkway ' x '.

OPTIONS: (A) $(12 + 2x)(10 + 2x) = 360$, $x = 4$ (B) $(12 + 2x)(10 + 2x) = 360$, $x = 5$ (C) $(12 + x)(10 + x) = 360$, $x = 4$ (D) $(12 + x)(10 + x) = 360$, $x = 5$

ANSWER: (A) $(12 + 2x)(10 + 2x) = 360$, $x = 4$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The perimeter of a rectangle is 70 cm. The length of the rectangle is 5 cm more than twice its breadth. Express the given situation as a system of linear equations in two variables and hence solve it.

OPTIONS: (A) Length = 25 cm, Breadth = 10 cm (B) Length = 30 cm, Breadth = 5 cm (C) Length = 20 cm, Breadth = 15 cm (D) Length = 35 cm, Breadth = 0 cm

ANSWER: (A) Length = 25 cm, Breadth = 10 cm

YEAR: 2025-06

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Express the equation $(x-2)/(x-3) + (x-4)/(x-5) = 10/3$; ($x \neq 3, 5$) as a quadratic equation in standard form. Hence, find the roots of the equation so formed.

OPTIONS: (A) $2x^2 - 19x + 42 = 0$, roots: 6, $7/2$ (B) $2x^2 - 19x + 42 = 0$, roots: 6, $7/2$ (C) $2x^2 - 19x + 42 = 0$, roots: 6, $7/2$ (D) $2x^2 - 19x + 42 = 0$, roots: 6, $7/2$

ANSWER: (A) $2x^2 - 19x + 42 = 0$, roots: 6, $7/2$

YEAR: 2025-06

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find the smallest value of p for which the quadratic equation $x^2 - 2(p + 1)x + p^2 = 0$ has real roots. Hence, find the roots of the equation so obtained.

OPTIONS: (A) $p = -1/2$, roots: $1/2, 1/2$ (B) $p = 1/2$, roots: $1/2, 1/2$ (C) $p = -1/2$, roots: $1, 1$ (D) $p = 1/2$, roots: $1, 1$

ANSWER: (A) $p = -1/2$, roots: $1/2, 1/2$

CHAPTER 5 - ARITHMETIC PROGRESSIONS

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CHAPTER: 5 - Arithmetic Progressions

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YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The 9th term from the end (towards first term) of the AP 7, 11, 15, 19, ..., 147 is:

OPTIONS: (A) 135 (B) 125 (C) 115 (D) 39

ANSWER: (C) 115

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Three numbers in AP have the sum 30. What is its middle term?

OPTIONS: (A) 4 (B) 10 (C) 16 (D) 8

ANSWER: (B) 10

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The 10th term of the AP 5, $19/4$, $9/2$, $17/4$, ... is:

OPTIONS: (A) $11/4$ (B) $4/11$ (C) $13/4$ (D) $4/13$

ANSWER: (A) $11/4$

YEAR: 2025-06

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Cable cars at hill stations are one of the major tourist attractions. On a hill station, the length of cable car ride from base point to top most point on the hill is 5000 m. The distance of first pole from base point is 200 m and subsequent poles are installed at equal interval of 150 m. Further, the distance of last pole from the top is 300 m. Find the distance of 10th pole from the base. Find the distance between 15th pole and 25th pole. Find the time taken by cable car to reach 15th pole from the top if it is moving at the speed of 5 m/sec and coming from top.

OPTIONS: (A) 1550 m, 1500 m, 480 seconds (B) 1550 m, 1500 m, 480 seconds (C) 1550 m, 1500 m, 480 seconds (D) 1550 m, 1500 m, 480 seconds

ANSWER: (A) 1550 m, 1500 m, 480 seconds

CHAPTER 6 - COORDINATE GEOMETRY

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CHAPTER: 6 - Coordinate Geometry

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YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The coordinates of the end points of a diameter of a circle are (5, -2) and (5, 2). The length of the radius of the circle is:

OPTIONS: (A) ± 2 (B) ± 4 (C) 4 (D) 2

ANSWER: (D) 2

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The points (-5, 0), (5, 0) and (0, 4) are the vertices of a triangle which is a/an:

OPTIONS: (A) right-angled triangle (B) isosceles triangle (C) equilateral triangle (D) scalene triangle

ANSWER: (B) isosceles triangle

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The line represented by the equation $x - y = 0$ is:

OPTIONS: (A) parallel to x-axis (B) parallel to y-axis (C) passing through the origin (D) passing through the point (3, 2)

ANSWER: (C) passing through the origin

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The line represented by $x/4 + y/6 = 1$, intersects x-axis and y-axis respectively at P and Q. The coordinates of the mid-point of line segment PQ are:

OPTIONS: (A) (2, 3) (B) (3, 2) (C) (2, 0) (D) (0, 3)

ANSWER: (A) (2, 3)

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two of the vertices of ΔPQR are $P(-1, 5)$ and $Q(5, 2)$. The coordinates of a point which divides PQ in the ratio 2 : 1 are:

OPTIONS: (A) (3, -3) (B) (5, 5) (C) (3, 3) (D) (5, 1)

ANSWER: (C) (3, 3)

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: AOBC is a rectangle whose three vertices are $A(0, 2)$, $O(0, 0)$ and $B(4, 0)$. The square of the length of its diagonal is equal to:

OPTIONS: (A) 36 (B) 20 (C) 16 (D) 4

ANSWER: (B) 20

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of the point (4, 0) from x-axis is:

OPTIONS: (A) 4 units (B) 16 units (C) 0 units (D) $4\sqrt{2}$ units

ANSWER: (C) 0 units

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The equation of a line parallel to x-axis and at a distance of 3 units below x-axis is:

OPTIONS: (A) $x = 3$ (B) $x = -3$ (C) $y = -3$ (D) $y = 3$

ANSWER: (C) $y = -3$

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of point (a, -b) from x-axis is:

OPTIONS: (A) a (B) -a (C) b (D) -b

ANSWER: (C) b

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of point P(3a, 4a) from y-axis is:

OPTIONS: (A) 3a (B) -3a (C) 4a (D) -4a

ANSWER: (A) 3a

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The coordinates of the centre of a circle are $(2a, a - 7)$. Find the value(s) of 'a' if the circle passes through the point $(11, -9)$ and has diameter $10\sqrt{2}$ units.

OPTIONS: (A) 5, 3 (B) 4, 2 (C) 6, 4 (D) 7, 5

ANSWER: (A) 5, 3

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the length of the median through the vertex B of $\triangle ABC$ with vertices $A(9, -2)$, $B(-3, 7)$ and $C(-1, 10)$.

OPTIONS: (A) $\sqrt{58}$ (B) $\sqrt{60}$ (C) $\sqrt{55}$ (D) $\sqrt{50}$

ANSWER: (A) $\sqrt{58}$

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

OPTIONS: (A) $P(-1, 7/2)$, $Q(0, 5)$, $R(1, 13/2)$ (B) $P(-1, 7/2)$, $Q(0, 5)$, $R(1, 13/2)$ (C) $P(-1, 7/2)$, $Q(0, 5)$, $R(1, 13/2)$ (D) $P(-1, 7/2)$, $Q(0, 5)$, $R(1, 13/2)$

ANSWER: (A) $P(-1, 7/2)$, $Q(0, 5)$, $R(1, 13/2)$

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If the points $A(6, 1)$, $B(p, 2)$, $C(9, 4)$ and $D(7, q)$ are the vertices of a parallelogram ABCD, then find the values of p and q. Hence, check whether ABCD is a rectangle or not.

OPTIONS: (A) $p = 8$, $q = 3$; Not a rectangle (B) $p = 8$, $q = 3$; Rectangle (C) $p = 7$, $q = 4$; Not a rectangle (D) $p = 7$, $q = 4$; Rectangle

ANSWER: (A) $p = 8$, $q = 3$; Not a rectangle

YEAR: 2025-06

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find a relation between x and y such that $P(x, y)$ is equidistant from the points $A(3, 5)$ and $B(7, 1)$. Hence, write the coordinates of the points on x -axis and y -axis which are equidistant from points A and B .

OPTIONS: (A) $x - y = 2$; $(2, 0)$ and $(0, -2)$ (B) $x - y = 2$; $(2, 0)$ and $(0, -2)$ (C) $x - y = 2$; $(2, 0)$ and $(0, -2)$ (D) $x - y = 2$; $(2, 0)$ and $(0, -2)$

ANSWER: (A) $x - y = 2$; $(2, 0)$ and $(0, -2)$

CHAPTER 7 - TRIANGLES

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CHAPTER: 7 - Triangles

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YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In triangles ABC and DEF , $\angle B = \angle E$, $\angle F = \angle C$ and $AB = 3DE$. Then, the two triangles are:

OPTIONS: (A) congruent but not similar (B) congruent as well as similar (C) neither congruent nor similar (D) similar but not congruent

ANSWER: (D) similar but not congruent

YEAR: 2025-06

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the adjoining figure, $PQ \parallel XY \parallel BC$, $AP = 2$ cm, $PX = 1.5$ cm and $BX = 4$ cm. If $QY = 0.75$ cm, then $AQ + CY =$

OPTIONS: (A) 6 cm (B) 4.5 cm (C) 3 cm (D) 5.25 cm

ANSWER: (C) 3 cm

YEAR: 2025-06

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Given $\triangle ABC \sim \triangle PQR$, $\angle A = 30^\circ$ and $\angle Q = 90^\circ$. The value of $(\angle R + \angle B)$ is

OPTIONS: (A) 90° (B) 120° (C) 150° (D) 180°

ANSWER: (C) 150°

YEAR: 2025-06

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, $AD/BD = AE/EC$ and $\angle BDE = \angle CED$, prove that $\triangle ABC$ is an isosceles triangle.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-06

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, $AP = 1$ cm, $BP = 2$ cm, $AQ = 1.5$ cm and $AC = 4.5$ cm. Prove that $\triangle APQ \sim \triangle ABC$. Hence find PQ , if $BC = 3.6$ cm.

OPTIONS: (A) $PQ = 1.2$ cm (B) $PQ = 1.5$ cm (C) $PQ = 1.8$ cm (D) $PQ = 2.0$ cm

ANSWER: (A) $PQ = 1.2$ cm

YEAR: 2025-06

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If a line drawn parallel to one side of triangle intersecting the other two sides in distinct points divides the two sides in the same ratio, then it is parallel to third side. State and prove the converse of the above statement.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

CHAPTER 8 - CIRCLES

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CHAPTER: 8 - Circles

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YEAR: 2025-06

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For a circle with centre O and radius 5 cm, which of the following statements is true?

OPTIONS: (A) Distance between every pair of parallel tangents is 5 cm. (B) Distance between every pair of parallel tangents is 10 cm. (C) Distance between every pair of parallel tangents must be between 5 cm and 10 cm. (D) There does not exist a point outside the circle from where length of tangent is 5 cm.

ANSWER: (B) Distance between every pair of parallel tangents is 10 cm.

YEAR: 2025-06

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the adjoining figure, PA and PB are tangents to a circle with centre O such that $\angle P = 90^\circ$. If $AB = 3\sqrt{2}$ cm, then the diameter of the circle is

OPTIONS: (A) $3\sqrt{2}$ cm (B) $6\sqrt{2}$ cm (C) 3 cm (D) 6 cm

ANSWER: (D) 6 cm

YEAR: 2025-06

CHAPTER: 8 - Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-06

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, PC is a tangent to the circle at C. AOB is the diameter which when extended meets the tangent at P. Find $\angle CBA$ and $\angle BCO$, if $\angle PCA = 110^\circ$.

OPTIONS: (A) $\angle CBA = 70^\circ$, $\angle BCO = 70^\circ$ (B) $\angle CBA = 70^\circ$, $\angle BCO = 70^\circ$ (C) $\angle CBA = 70^\circ$, $\angle BCO = 70^\circ$ (D) $\angle CBA = 70^\circ$, $\angle BCO = 70^\circ$

ANSWER: (A) $\angle CBA = 70^\circ$, $\angle BCO = 70^\circ$

CHAPTER 9 - INTRODUCTION TO TRIGONOMETRY

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CHAPTER: 9 - Introduction to Trigonometry

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YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $\tan^2\theta - (1/\cos\theta \times \sec\theta)$ is:

OPTIONS: (A) 1 (B) 0 (C) -1 (D) 2

ANSWER: (C) -1

YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sin\theta = \cos\theta$ ($0^\circ < \theta < 90^\circ$), then the value of $\sec\theta \cdot \sin\theta$ is:

OPTIONS: (A) $1/\sqrt{2}$ (B) $\sqrt{2}$ (C) 0 (D) 1

ANSWER: (D) 1

YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $(\cot\theta + \tan\theta)$ equals:

OPTIONS: (A) $\operatorname{cosec}\theta \sec\theta$ (B) $\sin\theta \sec\theta$ (C) $\cos\theta \tan\theta$ (D) $\sin\theta \cos\theta$

ANSWER: (A) $\operatorname{cosec}\theta \sec\theta$

YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\tan 30^\circ = \frac{1}{\sqrt{3}}$, then θ equals:

OPTIONS: (A) 60° (B) 30° (C) 20° (D) 10°

ANSWER: (D) 10°

YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(1 - 2 \sin^2 60^\circ)$ is same as that of

OPTIONS: (A) $\sin 30^\circ$ (B) $-\sin 30^\circ$ (C) $\cos 60^\circ$ (D) $-\cos 30^\circ$

ANSWER: (B) $-\sin 30^\circ$

YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sec A = 2 \cos A$ is true for $A =$

OPTIONS: (A) 0° (B) 30° (C) 45° (D) 60°

ANSWER: (C) 45°

YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(1 + \cot^2 \theta)(1 + \tan^2 \theta) = \frac{1}{(\sin^2 \theta - \sin^4 \theta)}$

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-06

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\tan A + \cot A = 6$, then find the value of $\tan^2 A + \cot^2 A - 4$.

OPTIONS: (A) 30 (B) 32 (C) 28 (D) 34

ANSWER: (A) 30

CHAPTER 10 - AREAS RELATED TO CIRCLES

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CHAPTER: 10 - Areas Related to Circles

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YEAR: 2025-06

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If a large circular pizza is divided into 5 equal sectors, then the central angle of each sector will be:

OPTIONS: (A) 60° (B) 90° (C) 45° (D) 72°

ANSWER: (D) 72°

YEAR: 2025-06

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If an arc of a circle of diameter 10 cm subtends an angle of 144° at the centre of the circle, then the length of the arc is:

OPTIONS: (A) 2π cm (B) 4π cm (C) 5π cm (D) 6π cm

ANSWER: (B) 4π cm

YEAR: 2025-06

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, the shape of the top of a table is that of a sector of a circle with centre O and $\angle AOB = 90^\circ$. If $AO = OB = 42$ cm, then find the perimeter of the top of the table.

OPTIONS: (A) 282 cm (B) 300 cm (C) 250 cm (D) 320 cm

ANSWER: (A) 282 cm

YEAR: 2025-06

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, three sectors of a circle of radius 5 cm, making angles 35° , 50° and 95° at the centre are shaded. Find the area of the shaded region. [Use $\pi = 22/7$]

OPTIONS: (A) $275/7 \text{ cm}^2$ (B) $250/7 \text{ cm}^2$ (C) $300/7 \text{ cm}^2$ (D) $325/7 \text{ cm}^2$

ANSWER: (A) $275/7 \text{ cm}^2$

CHAPTER 11 - SURFACE AREAS AND VOLUMES

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CHAPTER: 11 - Surface Areas and Volumes

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YEAR: 2025-06

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: On the top face of the wooden cube of side 7 cm, hemispherical depressions of radius 0.35 cm are to be formed by taking out the wood. The maximum number of depressions that can be formed is:

OPTIONS: (A) 400 (B) 100 (C) 20 (D) 10

ANSWER: (B) 100

YEAR: 2025-06

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cone of height 12 cm and slant height 13 cm is surmounted on a hemisphere having radius equal to that of cone. The entire height of the solid is:

OPTIONS: (A) 17 cm (B) 18 cm (C) 22 cm (D) 23 cm

ANSWER: (A) 17 cm

YEAR: 2025-06

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The radii 'r' of a sphere and that of the base of a cone are same. If their volumes are also same, then the height of the cone is:

OPTIONS: (A) r (B) 2r (C) 3r (D) 4r

ANSWER: (D) 4r

CHAPTER 12 - STATISTICS

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CHAPTER: 12 - Statistics

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YEAR: 2025-06

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mean of seven observations is 17. If the mean of the first four observations is 15 and that of the last four observations is 18, then the fourth observation is:

OPTIONS: (A) 14 (B) 13 (C) 12 (D) 10

ANSWER: (B) 13

YEAR: 2025-06

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The cumulative frequency for calculating median is obtained by adding the frequencies of all the:

OPTIONS: (A) classes up to the median class (B) classes following the median class (C) classes preceding the median class (D) all classes

ANSWER: (C) classes preceding the median class

YEAR: 2025-06

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If mode and median of given set of observations are 13 and 11 respectively, then the value of mean is:

OPTIONS: (A) 17 (B) 7 (C) 10 (D) 28

ANSWER: (C) 10

YEAR: 2025-06

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If mean and median of given set of observations are 10 and 11 respectively, then the value of mode is:

OPTIONS: (A) 10.5 (B) 8 (C) 13 (D) 21

ANSWER: (C) 13

CHAPTER 13 - PROBABILITY

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CHAPTER: 13 - Probability

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YEAR: 2025-06

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The probability of drawing an even prime number out of numbers from 1 to 30 is:

OPTIONS: (A) $\frac{1}{30}$ (B) $\frac{4}{15}$ (C) $\frac{7}{30}$ (D) 0

ANSWER: (A) $\frac{1}{30}$

YEAR: 2025-06

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two coins are tossed simultaneously. The probability of getting at least one head is:

OPTIONS: (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) 1

ANSWER: (C) $\frac{3}{4}$

YEAR: 2025-06

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If in a lottery, there are 10 prizes and 30 blanks, then the probability of winning a prize is:

OPTIONS: (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$

ANSWER: (A) $\frac{1}{4}$

YEAR: 2025-06

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a cricket match, a batsman hits the boundary 7 times out of the 42 balls he plays. The probability of his not hitting a boundary is:

OPTIONS: (A) $\frac{1}{7}$ (B) $\frac{2}{7}$ (C) $\frac{5}{6}$ (D) $\frac{1}{6}$

ANSWER: (C) $\frac{5}{6}$

YEAR: 2025-06

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A bag contains red balls and black balls in the ratio 3 : 7. A ball is drawn at random. The probability that ball so drawn is black in colour, is:

OPTIONS: (A) $\frac{3}{7}$ (B) 0.3 (C) 0.7 (D) $\frac{1}{7}$

ANSWER: (C) 0.7
